



Power BI Dashboards for Health and Sleep Analysis

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Plan

- Introduction
- Problem Statement
- Project Objectives
- Proposed Solution
- Data Preprocessing
- Project Data Model
- Dashboards
- Conclusion



Introduction

➤ Use of two data sources:

- **Health and sleep statistics:** contains 100 rows and 12 columns (User ID, Age, Sleep Quality, Bedtime, Wake-up Time, Daily Steps, Calories Burned, Physical Activity Level, Dietary Habits, Sleep Disorders, Medication Usage)
- **Sleep Health and Lifestyle Dataset:** contains 400 rows and 13 columns (gender, age, occupation, sleep duration, quality of sleep, physical activity level, stress levels, BMI category, blood pressure, heart rate, daily steps, and the presence or absence of sleep disorders).

Problem Statement

- Impact of Stress on Sleep Quality
- Identifying Stress Factors
- Long-Term Health Consequences
- Solutions for Reducing Stress and Improving Sleep



Project Objectives

- This project aims to identify key stress factors, assess their impact on sleep, and propose data-driven solutions for reducing stress and improving overall well-being



Proposed Solution

- Using **Power BI** to analyze **sleep and stress data**, identifying key patterns and relationships, and presenting insights through intuitive dashboards for better decision-making and overall well-being.



Data Preprocessing

1

**Remove
unnecessary
columns**

2

**Standardize
column
names**

3

**Normalize
text values**

4

**Group values
into
categories**

5

**Convert data
types**

Remove unnecessary columns

- **Sleep Health and Lifestyle**
Dataset: Blood Pressure
- **Health and sleep statistics:**
Bedtime, Wake-up Time,
Calories Burned, Dietary Habits,
Medication Usage



Standardize column names

Ensure consistency between the two datasets by aligning column names:

- **User ID** (Health and Sleep Statistics) → **Person ID** (Sleep Health and Lifestyle Dataset)
- **Quality of Sleep** (Sleep Health and Lifestyle Dataset) → **Sleep Quality** (Health and Sleep Statistics)
- **Sleep Disorders** (Health and Sleep Statistics) → **Sleep Disorder** (Sleep Health and Lifestyle Dataset)

Normalize text values

Ensure consistent text formatting across datasets

- BMI Category(**Normal Weight**) → **Normal** (Sleep Health and Lifestyle Dataset)
- Sleep Disorder(**Sleep Apnea, Insomnia**) → **Yes/No** (Sleep Health and Lifestyle Dataset)
- Physical Activity Level (**min/day**) → **Low/Medium/High** (Sleep Health and Lifestyle Dataset)
- Gender (**F/M**) → **Female/Male** (Health and Sleep Statistics Dataset)

Group values into categories

➤ Group Ages into Meaningful Categories

- **20-29** if Age < 30
- **30-39** if Age < 40
- **40-49** if Age < 50
- **50-59** if Age < 60
- **60+** if Age $\geq$ 60



Convert data types

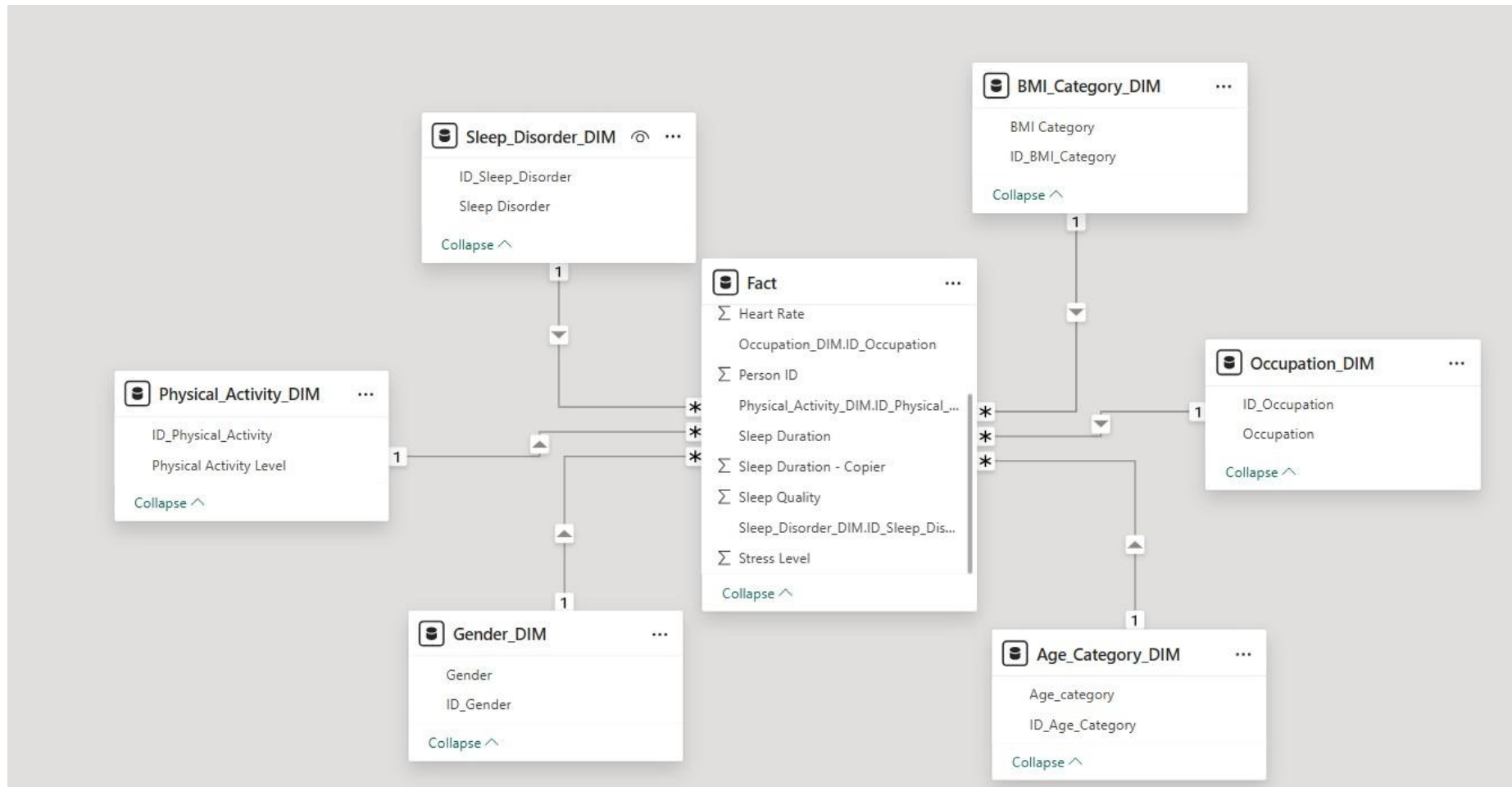
Convert Sleep Duration to Float (Sleep Health and Lifestyle Dataset)

- Convert the "**Sleep Duration**" column from a string format to a float for numerical analysis.



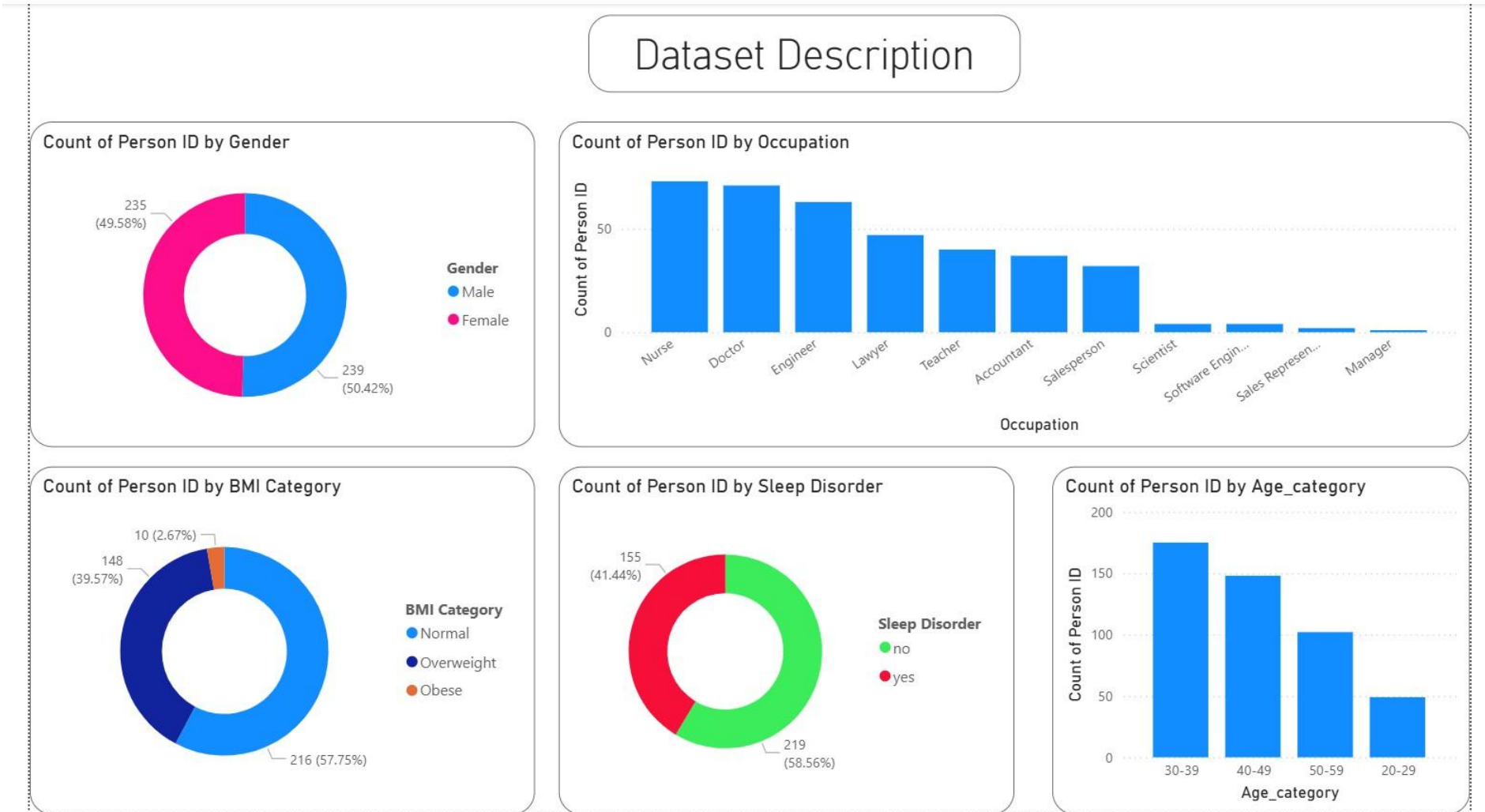
Project Data Model

- **Star Schema:** The two datasets were combined and structured into a dimensional model, featuring a central fact table and dimension tables for categorical attributes.



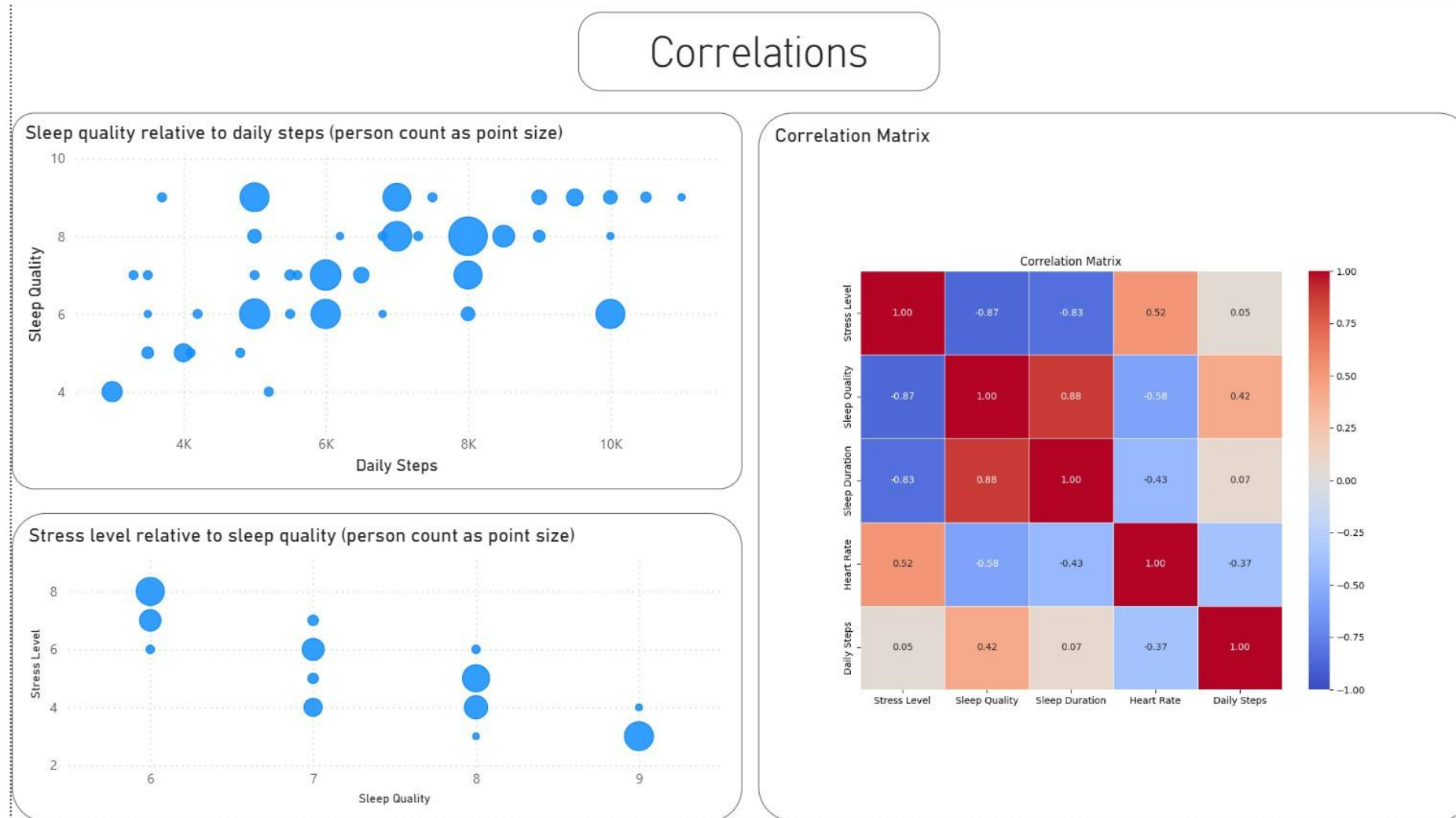
Dashboards

➤ Dataset Description dashboard



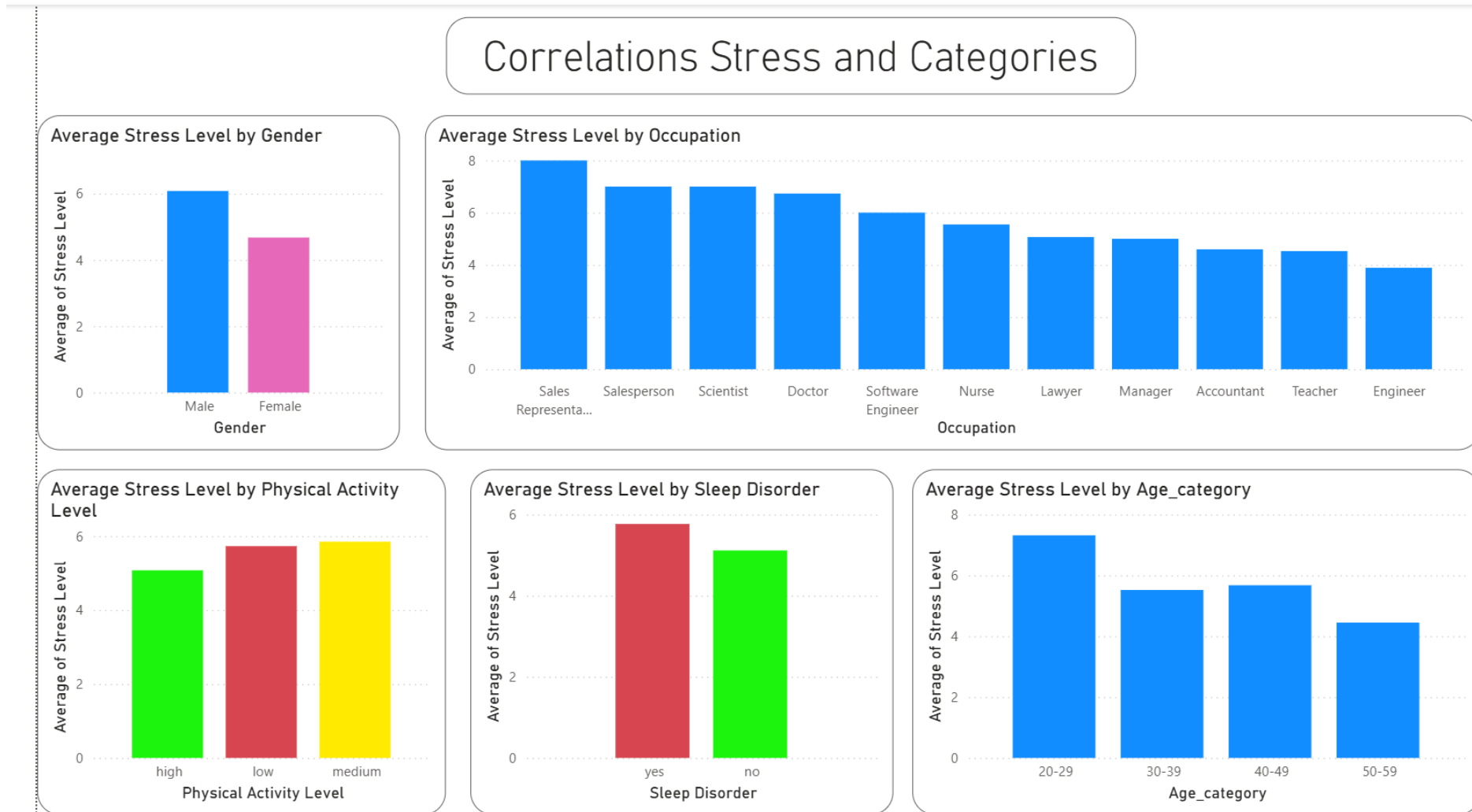
Dashboards

➤ Correlations dashboards



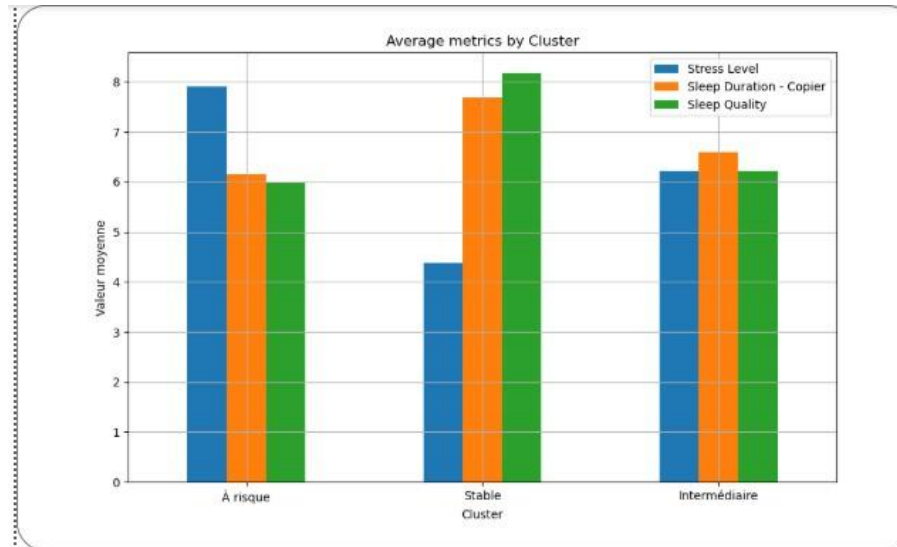
Dashboards

➤ Correlations Stress and Categories



Dashboards

➤ Clustering dashboard

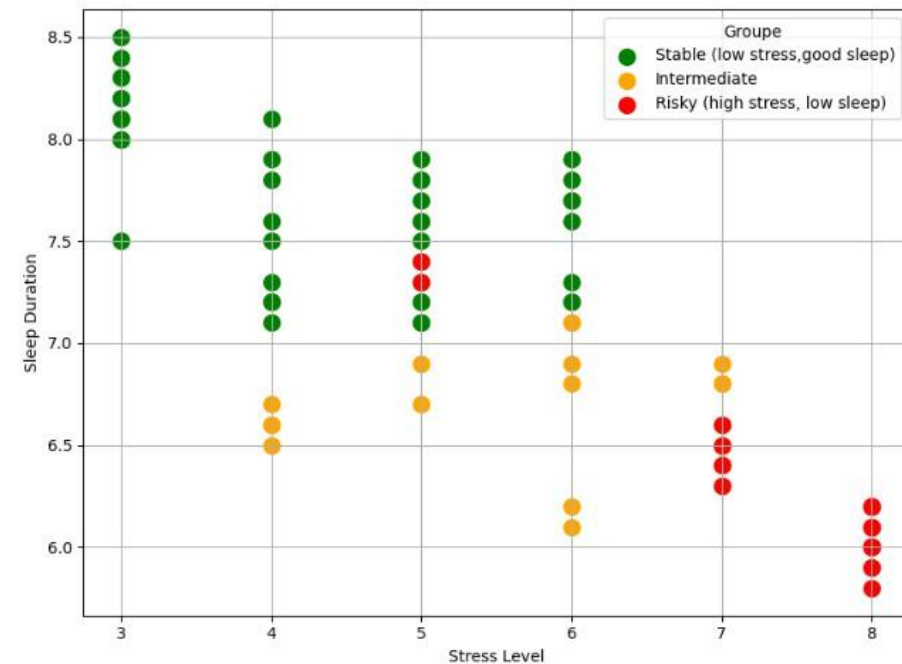


Dominant label per Cluster

Cluster	BMI dominant	Activity dominant
À risque	Overweight	medium
Stable	Normal	high
Intermédiaire	Normal	low

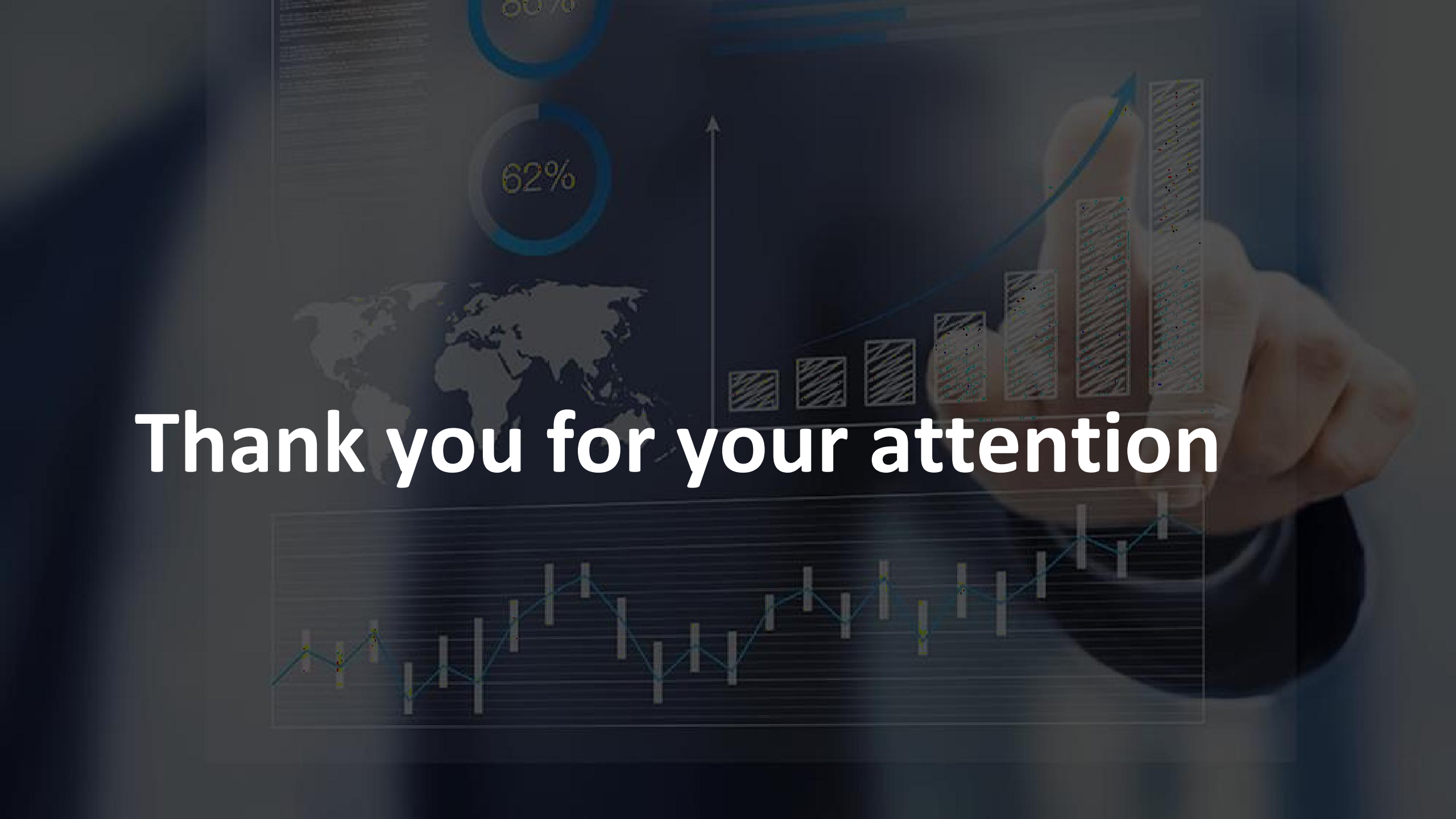
Clustering

Clustering relative to Stress Level, Sleep Quality, Sleep Duration - Copier, Heart Rate, Daily Steps, BMI Category, Physical Activity Level et Sleep Disorder





Conclusion



Thank you for your attention